# Dyadic-Comb Quadrature for the Fabius and Rvachev Functions

## Exact Polynomial Moments, Fractional Mellin Corrections, and Iterated Sums

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**Abstract**

Let $F$ be the Fabius function, let up be Rvachev's compactly supported $C^\infty$ function, and put $h = 2^{-m}$. This report studies weighted sums of $F$ and up on the dyadic comb $h\mathbb{Z}$, with particular emphasis on monomial weights. The main result is an exact shifted quadrature theorem:

$$h\sum_{k\in\mathbb{Z}}\big(h(k+\theta)\big)^p \operatorname{up}\big(h(k+\theta)\big) = \int_{-1}^{1} x^p \operatorname{up}(x)\,\mathrm{d}x \qquad (0 \le p \le m),$$

valid for every real shift $\theta$. Its mechanism is the increasing multiplicity of the integer zeros of the infinite sinc product $\widehat{\mathrm{up}}$. We derive the complete finite 2-adic alias decomposition, a closed first-failure series, phase-dependent superconvergence, and a Bell–Bernoulli formula for every higher alias jet.

For the one-sided Fabius comb we obtain a second exact stabilization law. If $M = 2^m$, $h = M^{-1}$, $p \in \mathbb{N}_0$, and $\tau(0) = 0$, $\tau(p) = 2\lceil p/2\rceil$ for $p \ge 1$, then

$$h\sum_{a=0}^{M}(ah)^p F(ah) = \frac{h^{p+1}B_{p+1}(M+1) - d_{p+1}}{p+1} \qquad (m \ge \tau(p)),$$

where $d_s$ is the Mellin moment of the Fabius probability density. An all-depth Hurwitz-zeta–Thue–Morse formula is also proved. For fractional powers, the punctured Rvachev comb has the universal singular correction $2\zeta(-\alpha)h^{\alpha+1}$, with a shifted version governed by two Hurwitz zeta values. Negative powers are treated by Mellin continuation and a logarithmic transition at $\alpha = -1$. Finally, the $n$-fold discrete prefix sum is reduced to a binomial Cauchy kernel and, at full support, to a finite combination of exact moments. The report includes exact-arithmetic and high-precision numerical experiments and records conjectures and directions for formalization, $q$-filters, inverse-Fabius sums, and Lambert-$W$ remainder asymptotics.


# Contents

## 1 Executive summary of the new results

The following statements are proved in this report. "New" below means new relative to the live ProveIt exposition and its consolidated frontier volume at the time of the audit; no universal priority claim is made without a broader literature search.

1. **Shift-independent exact quadrature for** $\mathrm{up}$**.** At depth $m$, every translate of the dyadic comb integrates exactly every polynomial of degree at most $m$ against $\mathrm{up}$.

2. **A finite** $2$**-adic alias hierarchy.** For a monomial of degree $p > m$, only the valuation layers $v_2(\ell) = 0, 1, \ldots, p-m-1$ can contribute to the Poisson error. Thus a formally infinite Fourier error has only finitely many arithmetic types.

3. **First-failure and superconvergent phases.** The degree-$m+1$ error is an explicit odd-frequency series involving $\widehat{\mathrm{up}}(q/2)$. If $m$ is even, the centered and half-shifted combs gain one extra exact odd degree. If $m$ is odd, the quarter-shifted combs gain one extra exact even degree.

4. **Bell–Bernoulli alias jets.** Every derivative of $\widehat{\mathrm{up}}$ at an integer zero is expressed by a complete exponential Bell polynomial. The finite sinc block contributes an explicit Bernoulli-number logarithmic jet.

5. **Exact Fabius right-comb stabilization.** For natural $p$, the right Riemann comb of $x^p F(x)$ becomes exactly equal to a finite Euler–Maclaurin polynomial as soon as $m \geq 2\lceil p/2 \rceil$. The closed form contains one Bernoulli polynomial and one Fabius-law moment.

6. **All-depth complex-power formula.** For arbitrary complex $\alpha$, the finite Fabius comb is a finite Thue–Morse sum of differences of Hurwitz zeta values. For natural $p$ this reduces to Bernoulli polynomials; for negative integers it reduces to harmonic-number combinations.

7. **Fractional and negative Rvachev powers.** For the punctured comb of $|x|^\alpha \,\mathrm{up}(x)$, the sole algebraic correction is $2\zeta(-\alpha)h^{\alpha+1}$; the rest is smaller than every power of $h$. The Mellin transform has only one pole, at $\alpha = -1$, because $\mathrm{up}(x) - 1$ is flat at the origin.

8. **Closed** $n$**-fold iterated sums.** The $n$-fold prefix sum has the exact binomial kernel $\binom{N-j+n-1}{n-1}$. At the right edge of the support, polynomial exactness converts it into a finite moment formula for $\mathrm{up}$ and a finite Bernoulli–moment formula for $F$.

9. **An integer-base extension.** The same proof works for the compact density whose Fourier transform is $\prod_{j\geq 0} \mathrm{sinc}(\pi\xi/b^j)$: the $b$-adic comb at depth $m$ is exact through degree $m$ for every integer $b \geq 2$.

# 2 Repository audit and separation from prior work

## 2.1 Scope of the audit

The current documentation tree contains the mathematical glossary, the main Fabius/Rvachev exposition, the non-elementarity article, the Lean walkthrough, the consolidated non-formalized frontier volume, a papers library, and a frozen archive of historical versions. The live TeX sources were read directly. The archive manifest was used to identify historical versions, byte-identical duplicates, and documents absorbed into the consolidated frontier volume; mathematically distinct archived themes were then cross-checked against that canonical consolidation. This avoids counting successive drafts as independent results while retaining their mathematical content.

The audit concentrated on the following existing ingredients:

- the relation between $F$ and up and the compact-support Fourier product;
- exact dyadic point values expressed through signed Thue–Morse sums and moments;
- the multiplicity and leading derivative of every integer zero of the infinite sinc product;
- Poisson summation and the unweighted partition of unity;
- moment recurrences, Bernoulli-polynomial identities, and $q$-product representations;
- continuous repeated integration and separate work on repeated prefix sums of the signed Thue–Morse sequence;
- endpoint and inverse-function asymptotics involving Lambert $W$.

Targeted searches in the live main exposition and consolidated frontier volume found no theorem formulated as a shifted dyadic-comb quadrature rule, no polynomial moment exactness statement of degree growing with the mesh depth, and no Fabius right-comb stabilization formula of the form proved below. The present report therefore treats those statements as new relative to the audited repository. Existing repeated-integration formulas are continuous Volterra identities; the iterated sums in Section 8 are discrete and use a different kernel.

## 2.2 Canonical repository sources

The live source tree is at

https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs.

The principal mathematical source is

https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex.

The canonical non-formalized frontier source is

https://github.com/VladimirReshetnikov/ProveIt/blob/main/Analysis/FabiusFunction/docs/non-formalized-research-frontiers/non-formalized-research-frontiers.tex.

Its README records the consolidation of the former dyadic, $q$-connection, repeated-integration, convergence, and inverse/saddle notebooks. The frozen archive and its provenance manifest are at https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/archive.

**Remark 2.1** (Novelty standard)**.** The exact identities below are proved from established ingredients, not inferred from numerical data. Nevertheless, a claim that they have never appeared anywhere in the literature would require a larger bibliographic search than was possible here. Conjectural statements are explicitly labeled.

# 3 Notation and established ingredients

## 3.1 Fabius and Rvachev normalizations

Let $\mathrm{up} : \mathbb{R} \to \mathbb{R}$ be the even $C^\infty$ function supported on $[-1, 1]$ and normalized by

$$\int_{-1}^{1} \mathrm{up}(x)\,\mathrm{d}x = 1.$$

On $[0, 1]$, the Fabius function and Rvachev function are related by

$$\mathrm{up}(x) = F(1-x), \qquad 0 \le x \le 1, \tag{1}$$

and on $[-1, 0]$ by $\mathrm{up}(x) = F(1 + x)$. Equivalently, the Fabius probability density on $[0, 1]$ is

$$\rho(x) := F'(x) = 2\,\mathrm{up}(2x-1). \tag{2}$$

Both $F$ at 0 and $1 - F$ at 1 are flat: every derivative vanishes at the corresponding endpoint. Consequently $\mathrm{up}(x) - 1$ is flat at $x = 0$, while up is flat at $x = \pm 1$.

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x)\mathrm{e}^{-2\pi \mathrm{i}\xi x}\,\mathrm{d}x.$$

The Fourier transform of up is the infinite sinc product

$$\Phi(\xi) := \widehat{\mathrm{up}}(\xi) = \prod_{j=0}^{\infty} \mathrm{sinc}\left(\frac{\pi\xi}{2^j}\right), \qquad \mathrm{sinc}\, z := \frac{\sin z}{z}. \tag{3}$$

It is even, real on the real axis, and rapidly decreasing together with all its derivatives.

## 3.2 Moments

Define centered moments

$$\mu_p := \int_{-1}^{1} x^p\,\mathrm{up}(x)\,\mathrm{d}x, \qquad c_n := \mu_{2n}. \tag{4}$$

Evenness gives $\mu_{2n+1} = 0$. The rational sequence $c_n$ satisfies

$$c_0 = 1, \qquad (2n+1)(2^{2n}-1)c_n = \sum_{j=0}^{n-1} \binom{2n+1}{2j} c_j. \tag{5}$$

Let

$$d_s := \int_0^1 x^s \rho(x)\,\mathrm{d}x \tag{6}$$

for complex $s$. Since $\rho$ is flat at both endpoints, $d_s$ is entire. For $r \in \mathbb{N}_0$,

$$d_r = 2^{-r} \sum_{j=0}^{\lfloor r/2 \rfloor} \binom{r}{2j} c_j. \tag{7}$$

Integration by parts gives the continuous Fabius moment

$$I_p := \int_0^1 x^p F(x)\,\mathrm{d}x = \frac{1 - d_{p+1}}{p+1}. \tag{8}$$

More generally, the right side extends to every complex exponent by a removable singularity at $p = -1$; see Section 7.6.

### 3.3 Integer zero multiplicities

For a nonzero integer $n$, write

$$d(n) := v_2(|n|) + 1, \qquad n = \sigma 2^{d(n)-1} q, \quad q \geq 1 \text{ odd}, \quad \sigma \in \{\pm 1\}.$$

Exactly $d(n)$ factors in (3) vanish at $n$, and the established integer-zero formula is

$$\operatorname{ord}_{\xi=n} \Phi = d(n), \qquad \Phi^{(d(n))}(n) = -\frac{d(n)!}{n^{d(n)}} \Phi\left(\frac{q}{2}\right). \tag{9}$$

This multiplicity growth is the central engine of the comb identities.

### 3.4 Exact dyadic point values

Put $\varepsilon_r = (-1)^{s_2(r)}$, where $s_2(r)$ is the binary digit sum, and define

$$P_m(t) := \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m}{2j} c_j t^{m-2j}, \qquad C_m := \frac{2^{-m(m+1)/2}}{m!}. \tag{10}$$

For $M = 2^m$ and $0 \leq a \leq M$, the exact point formula is

$$F\left(\frac{a}{M}\right) = C_m \sum_{r=0}^{a-1} \varepsilon_r P_m(2a - 2r - 1). \tag{11}$$

This formula supplies the finite arithmetic representations in Sections 5 and 6 and the exact numerical experiments.

## 4 Shifted dyadic-comb quadrature for up

Let

$$M = 2^m, \qquad h = M^{-1}, \qquad \theta \in \mathbb{R},$$

and define the shifted weighted comb

$$Q_{m,p}(\theta) := h \sum_{k \in \mathbb{Z}} \big(h(k+\theta)\big)^p \operatorname{up}\big(h(k+\theta)\big). \tag{12}$$

The sum is finite because up is compactly supported.

### 4.1 Exact Poisson representation

**Lemma 4.1** (Shifted Poisson formula)**.** *For every $p \in \mathbb{N}_0$,*

$$Q_{m,p}(\theta) = \mu_p + \left(-\frac{1}{2\pi\mathrm{i}}\right)^p \sum_{\ell\in\mathbb{Z}\setminus\{0\}} \mathrm{e}^{2\pi\mathrm{i}\ell\theta}\Phi^{(p)}(M\ell). \tag{13}$$

*The series is absolutely convergent.*

*Proof.* Apply shifted Poisson summation to $f_p(x) = x^p \,\mathrm{up}(x)$:

$$h\sum_k f_p(h(k+\theta)) = \sum_{\ell\in\mathbb{Z}} \mathrm{e}^{2\pi\mathrm{i}\ell\theta}\widehat{f}_p(M\ell).$$

Differentiating the Fourier transform gives

$$\widehat{f}_p(\xi) = \left(-\frac{1}{2\pi\mathrm{i}}\right)^p \Phi^{(p)}(\xi).$$

The zero-frequency term is $\mu_p$. Rapid decay of $\Phi^{(p)}$ gives absolute convergence. □

### 4.2 Exact polynomial moments

**Theorem 4.2** (Shifted dyadic exactness)**.** *For every $m \in \mathbb{N}_0$, every real shift $\theta$, and every polynomial $P$ of degree at most $m$,*

$$h\sum_{k\in\mathbb{Z}} P\big(h(k+\theta)\big)\,\mathrm{up}\big(h(k+\theta)\big) = \int_{-1}^{1} P(x)\,\mathrm{up}(x)\,\mathrm{d}x. \tag{14}$$

*In particular,*

$$Q_{m,p}(\theta) = \mu_p \qquad (0 \le p \le m). \tag{15}$$

*Proof.* For $\ell \neq 0$,

$$\mathrm{ord}_{\xi=M\ell}\,\Phi = v_2(M|\ell|) + 1 = m + v_2(|\ell|) + 1 \ge m+1.$$

Hence $\Phi^{(p)}(M\ell) = 0$ for every $p \le m$. The alias sum in (13) vanishes term by term. Linearity gives the polynomial statement. □

**Corollary 4.3** (Partition and exact centered moments)**.** *For every $m$ and $\theta$,*

$$h\sum_k \mathrm{up}(h(k+\theta)) = 1.$$

*For the centered comb,*

$$h\sum_k (kh)^{2r}\,\mathrm{up}(kh) = c_r \quad (2r \le m), \qquad h\sum_k (kh)^{2r+1}\,\mathrm{up}(kh) = 0.$$

*The second odd identity in fact holds at every depth by symmetry.*

**Remark 4.4** (Strang–Fix interpretation)**.** The theorem is a quadrature analogue of the Strang–Fix principle: zeros of a Fourier transform at the nonzero dual lattice, with increasing multiplicity, force polynomial reproduction. What is special here is that the generating function itself is fixed while the zero multiplicity sampled by the dyadic dual lattice grows linearly with $m$.

### 4.3 Finite 2-adic alias decomposition

For $p > m$, infinitely many aliases remain, but only finitely many valuation layers can be nonzero.

**Proposition 4.5** (Finite valuation hierarchy)**.** *For every $p \in \mathbb{N}_0$,*

$$Q_{m,p}(\theta) - \mu_p = \left(-\frac{1}{2\pi \mathrm{i}}\right)^p \sum_{v=0}^{p-m-1} \sum_{\substack{q\in\mathbb{Z} \\ q \text{ odd}}} \mathrm{e}^{2\pi \mathrm{i} 2^v q\theta} \Phi^{(p)}(2^{m+v}q), \tag{16}$$

*where the outer sum is empty when $p \le m$.*

*Proof.* Write every nonzero alias uniquely as $\ell = 2^v q$ with $q$ odd. Its zero order is $m + v + 1$. The $p$th derivative vanishes whenever $p < m + v + 1$, or equivalently $v > p - m - 1$. ☐

This decomposition is useful computationally: increasing $p$ activates one new 2-adic shell at a time. It also suggests valuation-sensitive Richardson filters; see Section 11.

### 4.4 The first failure and special phases

**Theorem 4.6** (Degree $m + 1$ error)**.** *For $p = m + 1$,*

$$\begin{aligned} Q_{m,m+1}(\theta) - \mu_{m+1} = &-\left(-\frac{1}{2\pi \mathrm{i}}\right)^{m+1} \frac{(m+1)!}{2^{m(m+1)}} \\ &\times \sum_{\substack{\ell\in\mathbb{Z} \\ \ell \text{ odd}}} \frac{\Phi(|\ell|/2)}{\ell^{m+1}} \mathrm{e}^{2\pi \mathrm{i}\ell\theta}. \end{aligned} \tag{17}$$

*The right side is not identically zero as a function of $\theta$. Thus the uniform degree $m$ in Theorem 4.2 is sharp.*

*Proof.* Only odd aliases survive by Theorem 4.5. At $M\ell$, the zero order is $m + 1$, and (9) gives

$$\Phi^{(m+1)}(M\ell) = -\frac{(m+1)!}{(M\ell)^{m+1}} \Phi(\ell/2).$$

Substitution into (13) yields (17). Its Fourier coefficient at frequency 1 is nonzero because $\Phi(1/2) \neq 0$. ☐

**Corollary 4.7** (Phase superconvergence)**.** *At the first failing degree $p = m + 1$:*

*(i) if $m$ is even, then $Q_{m,m+1}(0) = Q_{m,m+1}(1/2) = \mu_{m+1} = 0$;*

*(ii) if $m$ is odd, then $Q_{m,m+1}(1/4) = Q_{m,m+1}(3/4) = \mu_{m+1}$.*

*Proof.* If $m$ is even, then $m + 1$ is odd; pairing $\ell$ and $-\ell$ cancels at $\theta = 0$ and $1/2$. If $m$ is odd, then $m + 1$ is even; the pair is proportional to $\cos(2\pi\ell\theta)$, which vanishes for every odd $\ell$ at the quarter shifts. ☐

The complete zero set of the first-failure trigonometric series is not known; this becomes Theorem 11.2.

### 4.5 All higher integer jets: Bell and Bernoulli structure

The leading derivative (9) is enough for exactness, but arbitrary $p$ requires higher jets. They admit a compact factorization. For $n \neq 0$, write $d = d(n)$, $n = \sigma 2^{d-1} q$ as above, and define

$$P_d^{\mathrm{sinc}}(w) := \prod_{j=0}^{d-1} \operatorname{sinc}\left(\frac{\pi w}{2^j}\right). \tag{18}$$

Factoring the $d$ vanishing sinc terms gives the exact local identity

$$\Phi(n+w) = w^d U_n(w), \qquad U_n(w) = -\frac{P_d^{\mathrm{sinc}}(w)}{(n+w)^d} \Phi\left(\frac{q}{2} + \frac{\sigma w}{2^d}\right). \tag{19}$$

Let $Y_r$ denote the complete exponential Bell polynomial and set

$$\lambda_s(n) := \left. \frac{\mathrm{d}^s}{\mathrm{d}w^s} \log U_n(w) \right|_{w=0}.$$

**Proposition 4.8** (Bell-polynomial alias jets)**.** *For every $r \geq 0$,*

$$\Phi^{(d+r)}(n) = \frac{(d+r)!}{r!} U_n(0) Y_r(\lambda_1(n), \ldots, \lambda_r(n)), \qquad U_n(0) = -\frac{\Phi(q/2)}{n^d}. \tag{20}$$

*Moreover,*

$$\begin{aligned} \lambda_s(n) = &\left(\log P_d^{\mathrm{sinc}}\right)^{(s)}(0) + \sigma^s 2^{-ds} \left(\log \Phi\right)^{(s)}(q/2) \\ &+ (-1)^s \frac{d(s-1)!}{n^s}. \end{aligned} \tag{21}$$

*The finite sinc contribution vanishes for odd $s$, while for $s = 2j$,*

$$\left(\log P_d^{\mathrm{sinc}}\right)^{(2j)}(0) = -\frac{2^{2j-1}|B_{2j}|}{j} \pi^{2j} \frac{1 - 2^{-2jd}}{1 - 2^{-2j}}. \tag{22}$$

*Proof.* Differentiate $w^d U_n(w)$ and use $U_n^{(r)}(0) = U_n(0) Y_r(\lambda_1, \ldots, \lambda_r)$. Formula (21) follows by logarithmically differentiating (19). Finally,

$$\log \operatorname{sinc} z = -\sum_{j \geq 1} \frac{2^{2j-1}|B_{2j}|}{j(2j)!} z^{2j},$$

and summing the geometric progression over the $d$ sinc factors gives (22). $\square$

Equations (16) and (20) provide a fully algorithmic exact description for every natural degree $p$: only finitely many valuation layers are active, and every active derivative is a finite Bell polynomial in explicit local data.

# 5 Finite arithmetic formulas for centered and absolute Rvachev combs

The Fourier formulas are structurally transparent. For exact rational arithmetic at a fixed depth, it is useful to insert the point-value formula (11) directly.

Define the punctured absolute comb

$$A_{m,p} := h \sum_{k\in\mathbb{Z}\setminus\{0\}} |kh|^p \operatorname{up}(kh), \qquad p \in \mathbb{N}_0. \tag{23}$$

The puncture removes the ambiguous $0^0$ and is also the correct convention for negative powers. Since $\operatorname{up}(k/M) = F(1 - |k|/M)$,

$$A_{m,p} = \frac{2C_m}{M^{p+1}} \sum_{r=0}^{M-2} \varepsilon_r \sum_{t=0}^{M-r-2} (M-r-1-t)^p P_m(2t+1). \tag{24}$$

This is an exact finite rational formula for every $m, p \in \mathbb{N}_0$.

To expose Bernoulli polynomials, put $L = M - r - 1$. Expanding both powers and using Faulhaber's formula gives

$$\begin{aligned} A_{m,p} = \frac{2C_m}{M^{p+1}} \sum_{r=0}^{M-2} \varepsilon_r \sum_{j=0}^{\lfloor m/2\rfloor} \binom{m}{2j} c_j \sum_{u=0}^{p} (-1)^u \binom{p}{u} L^{p-u} \\ \times \sum_{v=0}^{m-2j} \binom{m-2j}{v} 2^v \frac{B_{u+v+1}(L) - B_{u+v+1}}{u+v+1}. \end{aligned} \tag{25}$$

For the signed centered comb,

$$Q_{m,p}(0) = \begin{cases} 0, & p \text{ odd}, \\ A_{m,p}, & p \text{ even and } p > 0, \\ 1, & p = 0. \end{cases}$$

When $p \le m$, (25) collapses to $c_{p/2}$ for even $p$ by Theorem 4.2. Thus the formula yields a family of nontrivial finite Thue–Morse–Bernoulli identities.

# 6 The one-sided Fabius dyadic comb

Define, first for natural $p$,

$$R_{m,p} := h \sum_{a=0}^{M} (ah)^p F(ah), \qquad M = 2^m, h = M^{-1}. \tag{26}$$

For $p = 0$ the term at $a = 0$ is zero because $F(0) = 0$.

## 6.1 All-depth formula for complex powers

For $\alpha \in \mathbb{C}$, define $0^\alpha F(0)$ by its continuous value 0. The flatness of $F$ at the origin makes this legitimate for every $\alpha$. Put

$$R_m(\alpha) := h \sum_{a=0}^{M} (ah)^\alpha F(ah). \tag{27}$$

**Theorem 6.1** (Hurwitz-zeta–Thue–Morse formula)**.** *For every $m \in \mathbb{N}_0$ and $\alpha \in \mathbb{C}$,*

$$\begin{aligned} R_m(\alpha) = \frac{C_m}{M^{\alpha+1}} \sum_{r=0}^{M-1} \varepsilon_r \sum_{j=0}^{\lfloor m/2\rfloor} \binom{m}{2j} c_j \sum_{\ell=0}^{m-2j} \binom{m-2j}{\ell} 2^\ell (-2r-1)^{m-2j-\ell} \\ \times \Big[\zeta(-\alpha-\ell, r+1) - \zeta(-\alpha-\ell, M+1)\Big]. \end{aligned} \tag{28}$$

*The difference of Hurwitz zeta functions is entire in $\alpha$; any apparent pole cancels.*

*Proof.* Insert (11) into (27), interchange the finite sums, and expand

$$(2a-2r-1)^{m-2j} = \sum_{\ell=0}^{m-2j} \binom{m-2j}{\ell} (2a)^{\ell} (-2r-1)^{m-2j-\ell}.$$

The remaining finite power sum is

$$\sum_{a=r+1}^{M} a^{\alpha+\ell} = \zeta(-\alpha-\ell, r+1) - \zeta(-\alpha-\ell, M+1).$$

□

For $\alpha = p \in \mathbb{N}_0$, use $\zeta(-n,x) = -B_{n+1}(x)/(n+1)$ to obtain a fully rational Bernoulli–Thue–Morse formula at every depth. For negative integer $\alpha$, the same expression becomes a finite combination of generalized harmonic numbers. Formula (28) is therefore the broadest closed form in this report: it includes natural, fractional, and negative powers without changing the finite outer arithmetic.

## 6.2 Exact stabilization for natural powers

Set

$$\tau(0) := 0, \qquad \tau(p) := 2\left\lceil \frac{p}{2} \right\rceil = \begin{cases} p, & p \text{ even}, \\ p+1, & p \text{ odd}, \end{cases} \qquad p \geq 1. \tag{29}$$

**Theorem 6.2** (Exact Fabius right-comb formula)**.** *For $p \in \mathbb{N}_0$ and $m \geq \tau(p)$,*

$$\boxed{R_{m,p} = \frac{h^{p+1} B_{p+1}(M+1) - d_{p+1}}{p+1}.} \tag{30}$$

*Equivalently,*

$$R_{m,p} = I_p + \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{\underline{2r-1}} h^{2r}. \tag{31}$$

*Thus the Euler–Maclaurin expansion terminates and becomes exact at the stated dyadic depth.*

*Proof.* Let $f_p(x) = x^p F(x)$ on $[0,1]$ and define the composite trapezoidal sum

$$T_{m,p} := h\left( \frac{f_p(0)}{2} + \sum_{a=1}^{M-1} f_p(ah) + \frac{f_p(1)}{2} \right).$$

Since $f_p(0) = 0$ and $f_p(1) = 1$,

$$R_{m,p} = T_{m,p} + \frac{h}{2}. \tag{32}$$

Poisson summation for the zero extension of $f_p$, with the standard half-value convention at the endpoints, gives

$$T_{m,p} = \sum_{\ell \in \mathbb{Z}} \widehat{f}_p(M\ell), \qquad \widehat{f}_p(\xi) = \int_0^1 f_p(x) \mathrm{e}^{-2\pi \mathrm{i} \xi x} \, \mathrm{d}x.$$

For a nonzero integer $q$, integrate by parts $p+1$ times. Flatness of $F$ at 0 and of $1-F$ at 1 gives

$$\widehat{f}_p(q) = -\sum_{s=0}^{p} \frac{p^{\underline{s}}}{(2\pi \mathrm{i} q)^{s+1}} + \frac{\widehat{g}_p(q)}{(2\pi \mathrm{i} q)^{p+1}}, \qquad g_p := D^{p+1}(x^p F). \tag{33}$$

Pairing $q$ and $-q$ in the first sum leaves exactly the Bernoulli terms in (31). It remains to show that the last alias sum vanishes.

Let $\rho = F'$ and $A = \widehat{\rho}$. The operator identity

$$g_p = \prod_{r=2}^{p+1} (xD + r)\rho \tag{34}$$

transforms into

$$\widehat{g}_p(\xi) = \prod_{s=1}^{p} (s - \xi D_\xi) A(\xi). \tag{35}$$

From (2),

$$A(\xi) = \mathrm{e}^{-\pi \mathrm{i} \xi} \Phi(\xi/2). \tag{36}$$

At $q = M\ell$, $\ell \neq 0$, the zero order of $A$ is

$$\operatorname{ord}_{\xi=q} A = m + v_2(|\ell|). \tag{37}$$

If $p$ is odd and $m \geq p+1$, this order is greater than $p$, so $\widehat{g}_p(q) = 0$ term by term. If $p$ is even and $m > p$, the same is true. At the boundary case $m = p$, only odd $\ell$ have zero order exactly $p$. Then every lower derivative vanishes and

$$\widehat{g}_p(q) = (-q)^p A^{(p)}(q).$$

Moreover, all lower derivatives of $\Phi(\xi/2)$ vanish at $q$, so

$$A^{(p)}(q) = \mathrm{e}^{-\pi \mathrm{i} q} 2^{-p} \Phi^{(p)}(q/2).$$

For even $p$ this is real and even in $q$ (including the case $p = m = 0$). Thus the numerator is even, whereas $(2\pi \mathrm{i} q)^{p+1}$ changes sign under $q \mapsto -q$. The two aliases cancel. Hence the remainder in (33) sums to zero whenever $m \geq \tau(p)$.

Finally, expanding $B_{p+1}(M+1)$ at $M = 1/h$ gives

$$\frac{h^{p+1} B_{p+1}(M+1) - 1}{p+1} = \frac{h}{2} + \sum_{r=1}^{\lfloor (p+1)/2 \rfloor} \frac{B_{2r}}{(2r)!} p^{\underline{2r-1}} h^{2r}.$$

Combining this with (8) proves (30). □

**Example 6.3.** The first cases are

$$R_{m,0} = \frac{1}{2} + \frac{h}{2},$$

$$R_{m,1} = \frac{1-d_2}{2} + \frac{h}{2} + \frac{h^2}{12} \qquad (m \geq 2),$$

$$R_{m,2} = \frac{1-d_3}{3} + \frac{h}{2} + \frac{h^2}{6} \qquad (m \geq 2),$$

and

$$R_{m,3} = \frac{1-d_4}{4} + \frac{h}{2} + \frac{h^2}{4} - \frac{h^4}{120} \qquad (m \geq 4).$$

**Remark 6.4.** The proof explains the parity in $\tau(p)$. Odd $p$ requires all aliases of $A$ to have order $> p$. Even $p$ permits the boundary order $p$ because the remaining Fourier terms cancel in symmetric pairs.

### 6.3 An exact absolute-power corollary for up

The Fabius result yields a stronger exact formula for the absolute centered comb than polynomial exactness alone.

**Theorem 6.5** (Absolute integer powers)**.** *For $p \in \mathbb{N}_0$ and $m \geq \tau(p)$,*

$$\boxed{A_{m,p} = \frac{2d_{p+1}}{p+1} + 2\zeta(-p)h^{p+1}.} \tag{38}$$

*The first term is the continuous absolute moment*

$$\int_{-1}^{1} |x|^p \operatorname{up}(x)\,\mathrm{d}x = \frac{2d_{p+1}}{p+1}. \tag{39}$$

*Consequently:*

- *for positive even $p$, the comb is exactly equal to the integral;*
- *for odd $p$, the sole error is the exact zeta correction $2\zeta(-p)h^{p+1}$;*
- *for $p = 0$, the punctured comb is $1 - h$, since $2\zeta(0) = -1$.*

*Proof.* For $p \geq 1$, use (1) and change $a = M - k$:

$$A_{m,p} = 2h\sum_{k=1}^{M}(kh)^p F(1 - kh) = 2\sum_{j=0}^{p}(-1)^j \binom{p}{j} R_{m,j}.$$

The same binomial combination of the integrals $I_j$ equals $2d_{p+1}/(p+1)$. Insert (31). Alternating finite differences annihilate every falling factorial of degree $< p$; the only surviving endpoint term occurs when $p$ is odd and has degree $p$. It equals $2\zeta(-p)h^{p+1}$ by $\zeta(-p) = -B_{p+1}/(p+1)$. The case $p = 0$ follows from the partition of unity after deleting the central term $h \operatorname{up}(0) = h$. □

This theorem is the integer resonance of the fractional-power expansion in the next section.

## 7 Fractional and negative powers

### 7.1 Branch conventions

On a two-sided comb, $x^\alpha$ is ambiguous for noninteger $\alpha$. We therefore separate

$$|x|^\alpha \qquad \text{and} \qquad \operatorname{sgn}(x)|x|^\alpha.$$

A chosen complex branch can be reconstructed. For example, with the principal value $(-x)^\alpha = \mathrm{e}^{\pi\mathrm{i}\alpha}x^\alpha$ for $x > 0$,

$$x^\alpha = \frac{1 + \mathrm{e}^{\pi\mathrm{i}\alpha}}{2}|x|^\alpha + \frac{1 - \mathrm{e}^{\pi\mathrm{i}\alpha}}{2}\operatorname{sgn}(x)|x|^\alpha. \tag{40}$$

### 7.2 The Mellin transform of up

For $\Re\alpha > -1$, define

$$\mathcal{M}_{\mathrm{up}}(\alpha) := \int_{-1}^{1} |x|^{\alpha}\,\mathrm{up}(x)\,\mathrm{d}x = 2\int_{0}^{1} x^{\alpha}\,\mathrm{up}(x)\,\mathrm{d}x. \tag{41}$$

Since $\mathrm{up}(x) - 1$ is flat at 0,

$$\mathcal{M}_{\mathrm{up}}(\alpha) = \frac{2}{\alpha+1} + 2\int_{0}^{1} x^{\alpha}(\mathrm{up}(x) - 1)\,\mathrm{d}x. \tag{42}$$

The second term is entire in $\alpha$. Therefore $\mathcal{M}_{\mathrm{up}}$ extends meromorphically to $\mathbb{C}$ with exactly one pole, at $\alpha = -1$, of residue 2.

Using $\mathrm{up}(x) = F(1-x)$ and the symmetric Fabius density gives the useful identity

$$\mathcal{M}_{\mathrm{up}}(\alpha) = \frac{2d_{\alpha+1}}{\alpha+1}, \tag{43}$$

first for $\Re\alpha > -1$ and then meromorphically. Thus the generalized Fabius moment $d_s$ is the natural analytic continuation of all absolute Rvachev moments.

### 7.3 Universal zeta correction

For every real $\alpha$, define the punctured comb

$$A_m(\alpha) := h \sum_{k\in\mathbb{Z}\setminus\{0\}} |kh|^{\alpha}\,\mathrm{up}(kh). \tag{44}$$

For $\alpha > -1$ this is a quadrature approximation to $\mathcal{M}_{\mathrm{up}}(\alpha)$.

**Lemma 7.1** (Cutoff-monomial summation)**.** *Let $\chi \in C_c^{\infty}([0,\infty))$ be equal to 1 near 0, let $0 < \theta \le 1$, and let $\beta > -1$. Then, for every $N > 0$,*

$$\begin{aligned} h\sum_{k=0}^{\infty}\big(h(k+\theta)\big)^{\beta}\chi\big(h(k+\theta)\big) = {} & \int_{0}^{\infty} x^{\beta}\chi(x)\,\mathrm{d}x \\ & + \zeta(-\beta,\theta)h^{\beta+1} + \mathcal{O}_{\beta,N,\chi}(h^{N}). \end{aligned} \tag{45}$$

*The formula continues meromorphically in $\beta$ if the integral is read as its Mellin finite part. The only collision of the two displayed residues is at $\beta = -1$.*

*Proof.* Put

$$\chi^*(s) := \int_{0}^{\infty} \chi(x)x^{s-1}\,\mathrm{d}x.$$

For $\Re s > 0$, integration by parts gives

$$\chi^*(s) = -\frac{1}{s}\int_{0}^{\infty} \chi'(x)x^{s}\,\mathrm{d}x.$$

The integral on the right is entire and rapidly decreasing in every vertical strip, because $\chi'$ is supported in a compact subinterval of $(0,\infty)$. Hence $\chi^*$ is meromorphic with a single pole at $s = 0$, of residue 1, and has rapid vertical decay away from that pole.

For $c > \beta + 1$, Mellin inversion and absolute convergence give

$$h^{\beta+1} \sum_{k=0}^{\infty} (k+\theta)^{\beta} \chi(h(k+\theta)) = \frac{1}{2\pi \mathrm{i}} \int_{c-\mathrm{i}\infty}^{c+\mathrm{i}\infty} \chi^*(s) \zeta(s-\beta, \theta) h^{\beta+1-s} \, \mathrm{d}s.$$

Move the contour to $\Re s = -A$. The Hurwitz zeta pole at $s = \beta + 1$ contributes $\chi^*(\beta + 1) = \int x^{\beta} \chi(x) \, \mathrm{d}x$, and the pole of $\chi^*$ at $s = 0$ contributes $\zeta(-\beta, \theta) h^{\beta+1}$. There are no other poles because $\chi$ is locally constant at the origin. Polynomial vertical-strip bounds for the Hurwitz zeta function and the rapid decay of $\chi^*$ make the shifted integral $\mathcal{O}(h^{\beta+1+A})$. Taking $A$ arbitrarily large proves (45). The same contour identity supplies the stated meromorphic continuation. □

**Theorem 7.2** (Fractional absolute-power asymptotic)**.** *Fix real $\alpha > -1$. For every $N > 0$,*

$$\boxed{A_m(\alpha) = \mathcal{M}_{\mathrm{up}}(\alpha) + 2\zeta(-\alpha) h^{\alpha+1} + \mathcal{O}_{\alpha,N}(h^N).} \tag{46}$$

*For nonnegative integer $\alpha = p$, the remainder becomes identically zero once $m \geq \tau(p)$, by Theorem 6.5.*

*Proof.* Choose a cutoff $\chi$ as in Theorem 7.1, supported inside the positive half of the support. On $[0, \infty)$ write

$$x^{\alpha} \operatorname{up}(x) = x^{\alpha} \chi(x) + x^{\alpha} (\operatorname{up}(x) - \chi(x)).$$

The second summand extends by zero to a $C_c^{\infty}$ function: flatness of up $-1$ removes the apparent singularity at 0, and flatness of up handles the support endpoint. Poisson summation therefore makes its comb error $\mathcal{O}(h^N)$ for every $N$. Apply Theorem 7.1 with $\theta = 1$ to the first summand. The positive half-comb contributes $\zeta(-\alpha) h^{\alpha+1}$, and evenness supplies an identical negative-half contribution. Adding the two integral terms gives $\mathcal{M}_{\mathrm{up}}(\alpha)$. □

The striking feature is the absence of an ordinary infinite Euler–Maclaurin series: the flatness of up $-1$ kills every higher local coefficient. At integer powers the zeta value explains the parity phenomenon: $\zeta(-2r) = 0$, while $\zeta(-(2r+1))$ is a nonzero Bernoulli number.

### 7.4 Shifted fractional combs

For $0 < \theta < 1$, define

$$A_{m,\theta}(\alpha) := h \sum_{k \in \mathbb{Z}} |h(k+\theta)|^{\alpha} \operatorname{up}(h(k+\theta)), \tag{47}$$

$$S_{m,\theta}(\alpha) := h \sum_{k \in \mathbb{Z}} \operatorname{sgn}(k+\theta) |h(k+\theta)|^{\alpha} \operatorname{up}(h(k+\theta)). \tag{48}$$

**Theorem 7.3** (Shifted Hurwitz corrections)**.** *For fixed real $\alpha > -1$, fixed $0 < \theta < 1$, and every $N > 0$,*

$$A_{m,\theta}(\alpha) = \mathcal{M}_{\mathrm{up}}(\alpha) + h^{\alpha+1} [\zeta(-\alpha, \theta) + \zeta(-\alpha, 1-\theta)] + \mathcal{O}(h^N), \tag{49}$$

$$S_{m,\theta}(\alpha) = h^{\alpha+1} [\zeta(-\alpha, \theta) - \zeta(-\alpha, 1-\theta)] + \mathcal{O}(h^N). \tag{50}$$

*Proof.* Split the shifted lattice at the origin. Its positive distances are $h(k+\theta)$ for $k \geq 0$, while its negative distances are $h(k+1-\theta)$ for $k \geq 0$. Apply Theorem 7.1 with shifts $\theta$ and $1-\theta$ to the two local monomials, and use Poisson summation for the two smooth flat remainders. The absolute comb adds the two singular contributions. The signed comb subtracts them, and its continuous integral vanishes because up is even. □

The two Hurwitz zeta values are the right- and left-hand singular lattice contributions. At natural integer powers, the reflection identity for Bernoulli polynomials makes these formulas consistent with the exact shifted quadrature theorem and the special phases in Theorem 4.7.

### 7.5 Negative powers and the logarithmic transition

For $-1 < \alpha < 0$, (46) is an ordinary convergent quadrature statement. At and below $-1$, the continuous integral diverges, but the Mellin continuation gives a canonical renormalization.

**Proposition 7.4** (Renormalized negative powers)**.** *For fixed real $\alpha < -1$, define the meromorphic value of $\mathcal{M}_{\mathrm{up}}(\alpha)$ by (42). Then, away from $\alpha = -1$,*

$$A_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1} \longrightarrow \mathcal{M}_{\mathrm{up}}(\alpha) \qquad (m \to \infty), \tag{51}$$

*with a superalgebraic remainder after the singular term is removed. At $\alpha = -1$,*

$$A_m(-1) = 2\log M + 2\gamma + 2\int_0^1 \frac{\mathrm{up}(x) - 1}{x}\,\mathrm{d}x + \mathcal{O}_N(h^N) \qquad \text{for every } N > 0. \tag{52}$$

*Proof.* Use the meromorphic continuation in Theorem 7.1 for the singular cutoff term and Poisson summation for the flat remainder. This proves (51), in fact with an $\mathcal{O}_{\alpha,N}(h^N)$ remainder for every $N$.

For the transition point write $\alpha = -1 + \varepsilon$. From (42),

$$\mathcal{M}_{\mathrm{up}}(-1+\varepsilon) = \frac{2}{\varepsilon} + 2\int_0^1 \frac{\mathrm{up}(x) - 1}{x}\,\mathrm{d}x + O(\varepsilon),$$

whereas

$$2\zeta(1-\varepsilon)h^{\varepsilon} = -\frac{2}{\varepsilon} + 2\log M + 2\gamma + O(\varepsilon).$$

The poles cancel. Taking the finite part at $\varepsilon = 0$ gives (52); the contour remainder is again smaller than every power of $h$. □

Thus:

- $-1 < \alpha < 0$: ordinary convergence after a decaying correction;
- $\alpha = -1$: logarithmic divergence;
- $\alpha < -1$: power divergence $2\zeta(-\alpha)h^{\alpha+1}$ plus a finite-part limit.

### 7.6 Fractional and negative powers for $F$

The one-sided Fabius problem is qualitatively simpler at the origin because $F$ is flat there. Define

$$I_F(\alpha) := \int_0^1 x^{\alpha} F(x)\,\mathrm{d}x. \tag{53}$$

This is an entire function of $\alpha$. Integration by parts gives

$$I_F(\alpha) = \frac{1 - d_{\alpha+1}}{\alpha+1}, \tag{54}$$

with a removable singularity at $\alpha = -1$. In particular,

$$I_F(-1) = -d_0' = -\int_0^1 \log x \, \rho(x) \, \mathrm{d}x. \tag{55}$$

Every negative power is integrable because $F(x) = O(x^N)$ for every $N$ as $x \downarrow 0$.

For the finite comb $R_m(\alpha)$, Theorem 6.1 is exact at every depth. Its large-$m$ Poincare expansion is especially simple.

**Theorem 7.5** (Fabius Euler–Maclaurin expansion)**.** *For fixed $\alpha \in \mathbb{C}$ and every $K \geq 1$,*

$$R_m(\alpha) = I_F(\alpha) + \frac{h}{2} + \sum_{r=1}^{K-1} \frac{B_{2r}}{(2r)!} \alpha^{\underline{2r-1}} h^{2r} + \mathcal{O}_{\alpha,K}(h^{2K}). \tag{56}$$

*For $\alpha = p \in \mathbb{N}_0$, the falling factorials terminate and Theorem 6.2 shows that the finite series is exactly equal to the comb once $m \geq \tau(p)$.*

*Proof.* The function $x^\alpha F(x)$, extended by zero at 0, is $C^\infty$. At $x = 0$ all derivatives vanish. At $x = 1$, flatness of $1 - F$ implies that its $n$th derivative equals $\alpha^{\underline{n}}$. The ordinary Euler–Maclaurin formula therefore reduces to (56); adding $h/2$ converts the trapezoidal sum to the endpoint-inclusive right comb. □

# 8 Iterated sums of arbitrary order

## 8.1 The universal binomial kernel

Let $a_j$ be a sequence indexed from a fixed lower bound $j_0$, and define the scaled prefix operator

$$(\mathsf{P}_h a)_N := h \sum_{j=j_0}^{N} a_j.$$

**Lemma 8.1** (Discrete Cauchy kernel)**.** *For every order $n \geq 1$,*

$$(\mathsf{P}_h^n a)_N = h^n \sum_{j=j_0}^{N} \binom{N - j + n - 1}{n - 1} a_j. \tag{57}$$

*Proof.* The formula is immediate for $n = 1$. Summing once more and using the hockey stick identity

$$\sum_{r=j}^{N} \binom{r - j + n - 1}{n - 1} = \binom{N - j + n}{n}$$

proves the induction step. □

This answers the purely combinatorial part of the iterated-sum problem. The Fabius/Rvachev structure enters when the upper endpoint is the right edge of the support.

### 8.2 Full-support $n$-fold Rvachev sum

Let

$$T^{(n)}_{m,p} := h^n \sum_{j=-M}^{M} \binom{M-j+n-1}{n-1} (jh)^p \operatorname{up}(jh). \tag{58}$$

Define the polynomial kernel

$$K_{n,h}(x) := \frac{1}{(n-1)!} \prod_{s=1}^{n-1} (1-x+sh). \tag{59}$$

Then

$$T^{(n)}_{m,p} = h \sum_{j=-M}^{M} (jh)^p K_{n,h}(jh) \operatorname{up}(jh). \tag{60}$$

The weight has degree $p+n-1$, so Theorem 4.2 applies.

**Theorem 8.2** (Exact $n$-fold Rvachev sum)**.** *If $m \geq p+n-1$, then*

$$\boxed{T^{(n)}_{m,p} = \int_{-1}^{1} x^p K_{n,h}(x) \operatorname{up}(x)\,\mathrm{d}x.} \tag{61}$$

*Equivalently, if $e_r$ denotes the elementary symmetric polynomial,*

$$T^{(n)}_{m,p} = \frac{1}{(n-1)!} \sum_{\ell=0}^{n-1} (-1)^{\ell} e_{n-1-\ell}(1+h, 1+2h, \ldots, 1+(n-1)h)\mu_{p+\ell}. \tag{62}$$

*Proof.* Equation (60) follows from

$$h^{n-1}\binom{M-j+n-1}{n-1} = \frac{1}{(n-1)!} \prod_{s=1}^{n-1} (1-jh+sh).$$

Apply polynomial exactness, then expand the product in powers of $x$. □

Because odd moments vanish, (62) usually contains only about half of its nominal terms. It is a complete closed form in rational moments $c_r$.

### 8.3 Full-support $n$-fold Fabius sum

Define

$$S^{(n)}_{m,p} := h^n \sum_{j=0}^{M} \binom{M-j+n-1}{n-1} (jh)^p F(jh). \tag{63}$$

As before,

$$S^{(n)}_{m,p} = h \sum_{j=0}^{M} (jh)^p K_{n,h}(jh) F(jh).$$

**Theorem 8.3** (Exact $n$-fold Fabius sum)**.** *If*

$$m \geq \tau(p+n-1) = 2\left\lceil \frac{p+n-1}{2} \right\rceil, \tag{64}$$

*then*

$$\boxed{\begin{aligned} S^{(n)}_{m,p} &= \frac{1}{(n-1)!}\sum_{\ell=0}^{n-1}(-1)^{\ell} e_{n-1-\ell}(1+h,\dots,1+(n-1)h) \\ &\quad\times \frac{h^{p+\ell+1}B_{p+\ell+1}(M+1)-d_{p+\ell+1}}{p+\ell+1}. \end{aligned}} \tag{65}$$

*Proof.* Expand $K_{n,h}$ and apply Theorem 6.2 to each power $p+\ell$. The threshold in (64) is the maximum of $\tau(p+\ell)$ for $0 \le \ell \le n-1$. □

### 8.4 Arbitrary upper endpoints and noninteger weights

At an intermediate dyadic endpoint $N = a < M$, Theorem 8.1 remains an exact answer:

$$h^n \sum_{j=0}^{a} \binom{a-j+n-1}{n-1} (jh)^p F(jh).$$

For natural $p$, inserting (11) and expanding the polynomial kernel reduces the result to finite Bernoulli-polynomial sums. For complex $\alpha$, the same manipulation produces finite differences of Hurwitz zeta functions, exactly as in Theorem 6.1.

For up with fractional or negative absolute powers, the kernel $K_{n,h}(x)$ has a Taylor series at the singular point $x = 0$. Applying the shifted singular Euler–Maclaurin theorem term by term yields corrections

$$\sum_{r\ge 0} \kappa_r \zeta(-\alpha - r) h^{\alpha+r+1},$$

where $\kappa_r$ is the $r$th Taylor coefficient of the even part of $K_{n,h}$. Unlike the unweighted case, more than one zeta term can occur because the iterated kernel is not locally constant.

## 9 Integer-base extension and $q$-connections

For an integer $b \ge 2$, consider

$$\Phi_b(\xi) := \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{\pi\xi}{b^j}\right). \tag{66}$$

It is the Fourier transform of a compactly supported $C^\infty$ probability density $\mathrm{up}_b$, obtained as an infinite convolution of centered uniform laws at geometrically decreasing scales.

**Theorem 9.1** ($b$-adic comb exactness)**.** *Let $h = b^{-m}$. For every real shift $\theta$ and every polynomial $P$ of degree at most $m$,*

$$h \sum_{k\in\mathbb{Z}} P(h(k+\theta))\, \mathrm{up}_b(h(k+\theta)) = \int_{\mathbb{R}} P(x)\, \mathrm{up}_b(x)\, \mathrm{d}x. \tag{67}$$

*Proof.* At every nonzero dual-lattice point $b^m\ell$, the factors with $j = 0, 1, \dots, m$ all vanish. Thus $\Phi_b$ has zero order at least $m+1$. Shifted Poisson summation repeats the proof of Theorem 4.2. □

This extension clarifies which part of the phenomenon is genuinely dyadic: the exactness requires an integer geometric base so that the Fourier zeros align with the dual lattice. The finer valuation and Thue–Morse structure is special to $b = 2$.

The $q$-connections appear at several levels:

- the finite sinc block in (18) is a geometric product whose logarithmic jets are finite $q$-geometric sums;
- the moment generating function

$$\int \mathrm{e}^{tx}\,\mathrm{up}(x)\,\mathrm{d}x = \prod_{j=0}^{\infty} \frac{\sinh(t/2^{j+1})}{t/2^{j+1}}$$

  produces Bernoulli and Bell polynomials by logarithmic expansion;
- valuation-layer cancellation suggests Gaussian-binomial or $q$-difference filters that annihilate selected alias shells;
- for a nonintegral scale ratio $q^{-1}$, exact lattice alignment is lost, but finite linear combinations of nearby scales may restore prescribed orders of cancellation.

# 10 Numerical experiments

All rational tests were performed with Python's `Fraction`; no floating-point tolerance was used. The accompanying file `fabius_dyadic_comb_experiments.py` contains the exact point-value formula, moment recurrences, Bernoulli polynomials, all-depth comb formula, shifted tests, and iterated sums. The fractional tests use `mpmath` with 70 decimal digits.

## 10.1 Moments

The first centered moments are

| $n$ | $c_n = \int x^{2n}\,\mathrm{up}(x)\,\mathrm{d}x$ |
|---|---|
| 0 | 1 |
| 1 | 1/9 |
| 2 | 19/675 |
| 3 | 583/59535 |
| 4 | 132809/32531625 |
| 5 | 46840699/24405225075 |

The first Fabius-law moments are

$$d_0 = 1, \quad d_1 = \frac{1}{2}, \quad d_2 = \frac{5}{18}, \quad d_3 = \frac{1}{6}, \quad d_4 = \frac{143}{1350}, \quad d_5 = \frac{19}{270}.$$

## 10.2 Observed stabilization thresholds

For $0 \le p \le 10$, the first depth at which (30) holds exactly is:

| $p$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| first exact $m$ | 0 | 2 | 2 | 4 | 4 | 6 | 6 | 8 | 8 | 10 | 10 |
| $\tau(p)$ | 0 | 2 | 2 | 4 | 4 | 6 | 6 | 8 | 8 | 10 | 10 |

This supports the optimality conjecture in Theorem 11.1.

### 10.3 Shifted first-failure errors

The following exact errors illustrate Theorem 4.7.

| $m$ | $p = m + 1$ | shift $\theta$ | $Q_{m,p}(\theta) - \mu_p$ |
|---|---|---|---|
| 1 | 2 | 0 | 1/72 |
| 1 | 2 | 1/4 | 0 |
| 2 | 3 | 0 | 0 |
| 2 | 3 | 1/4 | −4643/11059200 |
| 3 | 4 | 0 | −23/5529600 |
| 3 | 4 | 1/4 | 0 |
| 4 | 5 | 0 | 0 |
| 4 | 5 | 1/4 | 4966069/383551316951040 |

Every tested shift was exactly correct for all degrees $0 \le p \le m$.

### 10.4 Fractional corrected sequences

The table displays

$$A_m(\alpha) - 2\zeta(-\alpha)h^{\alpha+1}.$$

Rapid stabilization is visible even at modest depths.

| $m$ | $\alpha = 1/2$ | $\alpha = 3/2$ | $\alpha = -1/2$ |
|---|---|---|---|
| 5 | 0.4913969866635627 | 0.1707208835823527 | 2.7852273786291746 |
| 6 | 0.4913969866528344 | 0.1707208835839721 | 2.7852273785097793 |
| 7 | 0.4913969866527990 | 0.1707208835839732 | 2.7852273785136532 |
| 8 | 0.4913969866527990 | 0.1707208835839732 | 2.7852273785136552 |
| 9 | 0.4913969866527990 | 0.1707208835839732 | 2.7852273785136552 |
| 10 | 0.4913969866527990 | 0.1707208835839732 | 2.7852273785136552 |

For $\alpha = 5/2$, the corrected value stabilizes to $0.0754523281224172153620023427\ldots$.

### 10.5 Reproduction instructions

From the archive directory, run

```
python fabius_dyadic_comb_experiments.py > numerical_results.txt
```

The script has no mandatory third-party dependency for exact tests. If `mpmath` is unavailable, use

```
python fabius_dyadic_comb_experiments.py --skip-fractional
```

## 11 Conjectures and frontier directions

**Conjecture 11.1** (Optimal Fabius threshold)**.** *For every $p \ge 1$, the condition $m \ge \tau(p)$ in Theorem 6.2 is best possible: the stabilized formula fails at $m = \tau(p) - 1$.*

This is verified exactly for $1 \le p \le 10$. A proof should reduce the first nonvanishing Fabius alias layer to a signed odd-frequency series and establish that its value cannot cancel.

**Conjecture 11.2** (Classification of first-failure phases)**.** *Modulo* 1*, the phase zeros forced by parity in Theorem 4.7 are the only zeros common to every depth of the corresponding parity:*

- *for even $m$, only $\theta = 0, 1/2$ are universally superconvergent at degree $m+1$;*
- *for odd $m$, only $\theta = 1/4, 3/4$ are universally superconvergent at degree $m+1$.*

*Individual depths may have accidental additional zeros.*

### 11.1 Promising research programs

**1. Valuation-selective extrapolation.** Because degree $p$ activates only finitely many shells $v_2(\ell) \le p - m - 1$, combine depths $m, m+1, \dots, m+r$ with coefficients that annihilate the first $r$ shells. The coefficients should have a natural $q$-binomial description with $q = 2^{-1}$ or $2^{-p}$. This may yield exact finite filters rather than merely asymptotic Richardson extrapolation.

**2. Denominator and 2-adic valuation laws.** Equations (25), (30), and (65) invite exact valuation analysis. Determine the reduced denominator of $R_{m,p}$ and of the iterated sums, and isolate the contributions of Bernoulli denominators, $m!$, and the Fabius moment denominators. Von Staudt–Clausen should interact nontrivially with the dyadic cancellations.

**3. Complete alias-kernel sign theory.** A sign or total-positivity theorem for the odd-frequency coefficients $\Phi((2r+1)/2)$ would settle threshold optimality and phase uniqueness. The cosine expansion of up and the signed Thue–Morse product may provide two complementary approaches.

**4. Fractional transseries and Lambert $W$.** After removing the zeta singular term, the remainder is superalgebraic. The repo's endpoint/small-argument asymptotics suggest that its logarithm is controlled by the same saddle geometry and Lambert-$W$ scales as the Fourier decay of $\Phi$. Derive an explicit beyond-all-orders expansion for

$$A_m(\alpha) - \mathcal{M}_{\mathrm{up}}(\alpha) - 2\zeta(-\alpha)h^{\alpha+1}$$

uniformly in $\alpha$.

**5. Inverse-Fabius combs.** Study

$$h \sum_k x_k^p F^{-1}(x_k), \qquad x_k = h(k+\theta),$$

where the inverse has logarithmic/Lambert-$W$ endpoint behavior. Singular Euler–Maclaurin should produce non-power corrections, and the exact inverse dyadic formulas in the repository may permit arithmetic special cases.

**6. Nonintegral geometric ratios.** For $\Phi_q(\xi) = \prod_{j\ge0} \mathrm{sinc}(\pi q^j \xi)$ with $q^{-1} \notin \mathbb{N}$, exact lattice alignment is lost. Search for finite multiscale filters whose Fourier symbols vanish to prescribed order at the relevant approximate aliases. Gaussian binomial coefficients are a natural candidate.

**7. Formalization.** The proof of Theorem 4.2 requires a Schwartz-class Poisson theorem, Fourier differentiation, and the already formalized integer zero multiplicity. The Fabius stabilization theorem additionally needs a piecewise-smooth Poisson/trapezoidal lemma and the operator identity (35). These are modular Lean targets and would fit the repository's promotion policy.

**8. Multidimensional tensor and radial variants.** Tensor products immediately give exact cubature on dyadic lattices with coordinatewise degree bounds. More interesting is a radial construction whose Fourier transform inherits valuation-dependent zero multiplicities on spherical lattice shells.

## 12 Conclusion

Dyadic point arithmetic and dyadic-comb quadrature are two faces of the same Fourier structure. The point formulas use signed Thue–Morse cancellation in physical space; the comb formulas use increasing zero multiplicity in frequency space. Their combination produces exact results that are stronger than ordinary numerical quadrature:

- arbitrary shifted combs reproduce all up-moments through degree $m$;
- natural Fabius combs stabilize to a finite Bernoulli expression at a sharp-looking parity threshold;
- fractional powers have a universal zeta correction and no further algebraic tail;
- negative powers admit a canonical Mellin finite part;
- repeated summation closes in a finite moment basis.

The resulting framework connects Poisson summation, 2-adic zero geometry, Strang–Fix reproduction, Bernoulli and Bell polynomials, Hurwitz zeta values, Thue–Morse arithmetic, and the Lambert-$W$ asymptotic frontier. It also provides several concrete formalization and computation targets.

## A Proof details for the Bernoulli boundary sum

Starting from (33), the boundary contribution to $T_{m,p} - I_p$ is

$$-\sum_{s=0}^{p} \frac{p^{\underline{s}}}{(2\pi \mathrm{i} M)^{s+1}} \sum_{\ell \neq 0} \ell^{-(s+1)}.$$

Odd powers of $\ell^{-1}$ cancel. Put $s = 2r - 1$. Then

$$-\frac{2\zeta(2r)}{(2\pi \mathrm{i})^{2r}} M^{-2r} p^{\underline{2r-1}} = \frac{B_{2r}}{(2r)!} p^{\underline{2r-1}} h^{2r},$$

using

$$\zeta(2r) = (-1)^{r+1} \frac{B_{2r}(2\pi)^{2r}}{2(2r)!}.$$

This gives precisely (31) before the spectral remainder is considered.

## B Low-order exact identities

For convenient reference:

$$\mu_0 = 1, \qquad \mu_1 = 0, \qquad \mu_2 = \frac{1}{9}, \qquad \mu_3 = 0, \qquad \mu_4 = \frac{19}{675},$$
$$\mathcal{M}_{\mathrm{up}}(0) = 1, \qquad \mathcal{M}_{\mathrm{up}}(1) = \frac{5}{18}, \qquad \mathcal{M}_{\mathrm{up}}(2) = \frac{1}{9}, \qquad \mathcal{M}_{\mathrm{up}}(3) = \frac{143}{2700}.$$

For the punctured absolute comb, once the threshold is reached,

$$
\begin{aligned}
A_{m,0} &= 1 - h, \\
A_{m,1} &= \frac{5}{18} - \frac{h^2}{6}, \\
A_{m,2} &= \frac{1}{9}, \\
A_{m,3} &= \frac{143}{2700} + \frac{h^4}{60}, \\
A_{m,4} &= \frac{19}{675}.
\end{aligned}
$$

Here $2\zeta(-1) = -1/6$ and $2\zeta(-3) = 1/60$.

For the first iterated Rvachev orders,

$$
\begin{aligned}
T^{(1)}_{m,p} &= \mu_p, \qquad m \ge p, \\
T^{(2)}_{m,p} &= (1+h)\mu_p - \mu_{p+1}, \qquad m \ge p+1, \\
T^{(3)}_{m,p} &= \frac{1}{2}\left((1+h)(1+2h)\mu_p - (2+3h)\mu_{p+1} + \mu_{p+2}\right), \\
&\qquad\qquad\qquad\qquad m \ge p+2.
\end{aligned}
$$

## C Source audit table

| Source group | Role in this report |
|---|---|
| `FabiusFunction_Mathematical_Glossary.tex` | Normalizations, notation, and cross-reference map. |
| `Fabius_Function_and_Rvachev_Up.tex` | Fourier product, moments, Poisson summation, integer zero multiplicities, exact dyadic point values, Bernoulli and $q$ identities. |
| `Non_Elementarity_of_the_Fabius_Function.tex` | Analytic/flatness context; no comb formula duplicated. |
| `fabius_lean_walkthrough.tex` | Formalization boundary and candidate theorem dependencies. |
| `non-formalized-research-frontiers.tex` | Consolidated dyadic, $q$, repeated-integration, convergence, and inverse/saddle research; used to avoid duplication and identify open proof obligations. |
| Archive manifest and frozen historical versions | Provenance and identification of earlier drafts, including the separate $K$-fold signed Thue–Morse summation note. These are distinguished from the weighted Fabius/Rvachev comb sums studied here. |
| Papers library | Arias de Reyna sources and historical background. |

## D Implementation notes

The exact code uses the recurrence (5) and the point formula (11). At a fixed depth, all $F(a/2^m)$ are computed as a convolution of the Thue–Morse signs with the odd samples of $P_m$. The implementation is intentionally transparent rather than asymptotically optimized. Exact depths through $m \approx 10$ are fast enough for the tables in this report.

The all-depth natural-power check uses

$$\sum_{a=r+1}^{M} a^n = \frac{B_{n+1}(M+1) - B_{n+1}(r+1)}{n+1},$$

which is the integer specialization of (28). Bernoulli numbers are generated exactly from their triangular recurrence, so the code does not require a computer algebra system.

## References

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[7] I. Navot, *A further extension of the Euler–Maclaurin summation formula*, Journal of Mathematics and Physics 41 (1962), 155–163. https://doi.org/10.1002/sapm1962411155.

[8] NIST Digital Library of Mathematical Functions, chapters on Bernoulli polynomials and the Hurwitz zeta function. https://dlmf.nist.gov/.

[9] V. Reshetnikov, *Fabius Function and Rvachev Up*, live TeX exposition in the ProveIt repository, accessed 28 August 2026. https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs.

[10] V. Reshetnikov, *Non-formalized Fabius research frontiers*, consolidated TeX volume in the ProveIt repository, accessed 28 August 2026.